

Milestone 2 CSA – Report

Group 15: Seppe Hermans and Martina De Angelis

AI tools have been used as a coding assistant for syntax and debugging.

Subsystem selection

The test was performed using TCLab FOPDT Regression, provided at:

<https://apmonitor.com/pdc/index.php/Main/TCLabRegression>

The two heaters ran in the same environment (ambient temperature, air flow...). Between the tests there was a minimum of 10 minutes to make sure the heat of Heater 1 didn't affect the results for Heater 2.

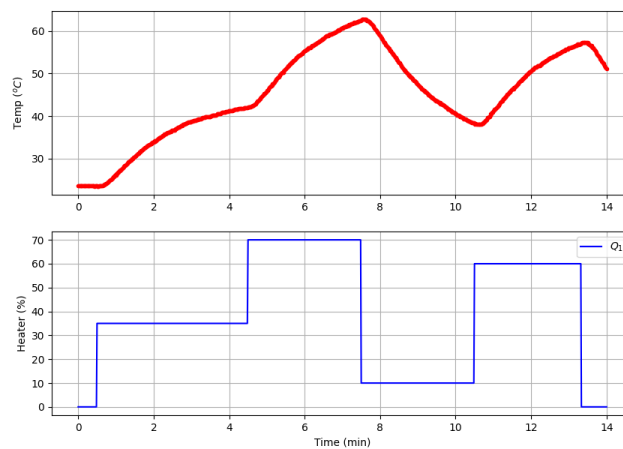


Figure 1: Heater 1 selection test.

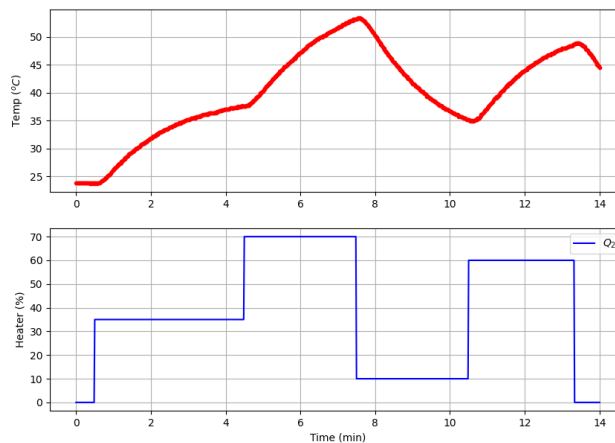


Figure 2: Heater 2 selection test.

H1:	Kp: 0.6596266349096659 taup: 125.18739435621976 thetap: 12.70999500254762	H2:	Kp: 0.49329427561595285 taup: 128.16026726026908 thetap: 11.949765199588892
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Conclusions

For the selection of the optimal subsystem between Heater 1 and Heater 2, the fundamental First-Order Plus Dead Time (FOPDT) parameters were used: steady-state gain K , time constant τ , and dead time θ . These parameters mathematically summarize the intrinsic thermal dynamics and the inherent controllability of each

plant. By quantitatively comparing these three empirical parameters, the selection was rigorously based on identifying the heater that offers the most favourable balance of dynamic responsiveness, control authority, and minimal latency.

Heater 1 presents higher gain, faster response speed, and lower dead time. Heater 1 is also found to have a lower noise standard deviation and lower peak-to-peak noise, which reinforces its suitability for control purposes. Repeatability tests instead favoured Heater 2. Despite this, H1 was retained as the working subsystem, since repeatability alone does not determine control performance: the faster dynamics and higher gain of H1 make it more responsive and easier to drive to a desired setpoint within the given time constraints. The results are presented below.

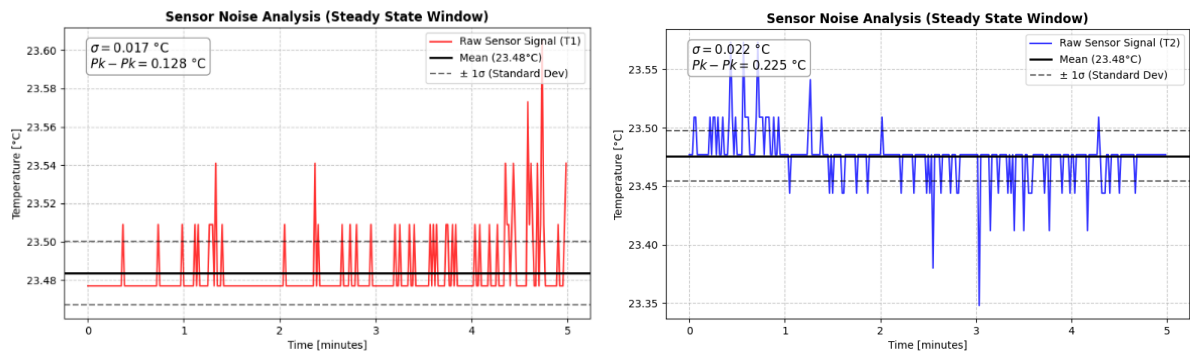


Figure 3 and 4: Noise analysis.

Heater H2 repeatability			
Phase	Repeatability [%]	RMSE [°C]	Max error [°C]
Heating 60%	98.015	0.505	1.096
Cooling 0%	97.412	0.534	0.967

Heater H1 repeatability			
Phase	Repeatability [%]	RMSE [°C]	Max error [°C]
Heating 60%	96.510	1.103	2.159
Cooling 0%	94.521	1.440	2.352

Heater 1 Model:

Transfer function: $G_1(s) = \frac{0.66 \cdot e^{-12.71s}}{125.19s+1}$

The selected heater behaves approximately as a first-order-plus-dead-time system. The gain $K = 0.6596$ means that for each 1% increase in heater power, the steady-state temperature increases by about 0.66°C. The time constant $\tau = 125.187$ shows that the system is slow, which is expected for a thermal system. The dead time $\theta = 12.71$ represents the delay before the sensor clearly reacts to a change in heater power.

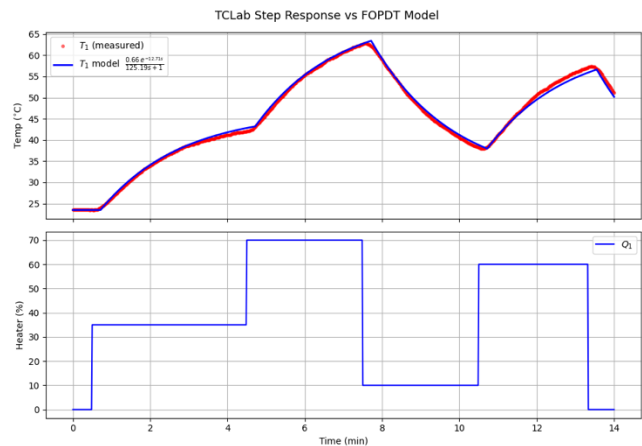


Figure 5: Model vs Real World.

Step 1 – Baseline controller

Baseline controller selection

For the baseline controller, a Proportional-Integral-Derivative (PID) architecture was selected. This controller combines proportional, integral, and derivative actions, making it suitable for evaluating the thermal response of the TCLab heater system under changing temperature setpoints. The code that we implemented, named *PID_heater.py*, origins from the apmonitor.com sample number 11 and has been adapted to give a live plot and appropriate tuning.

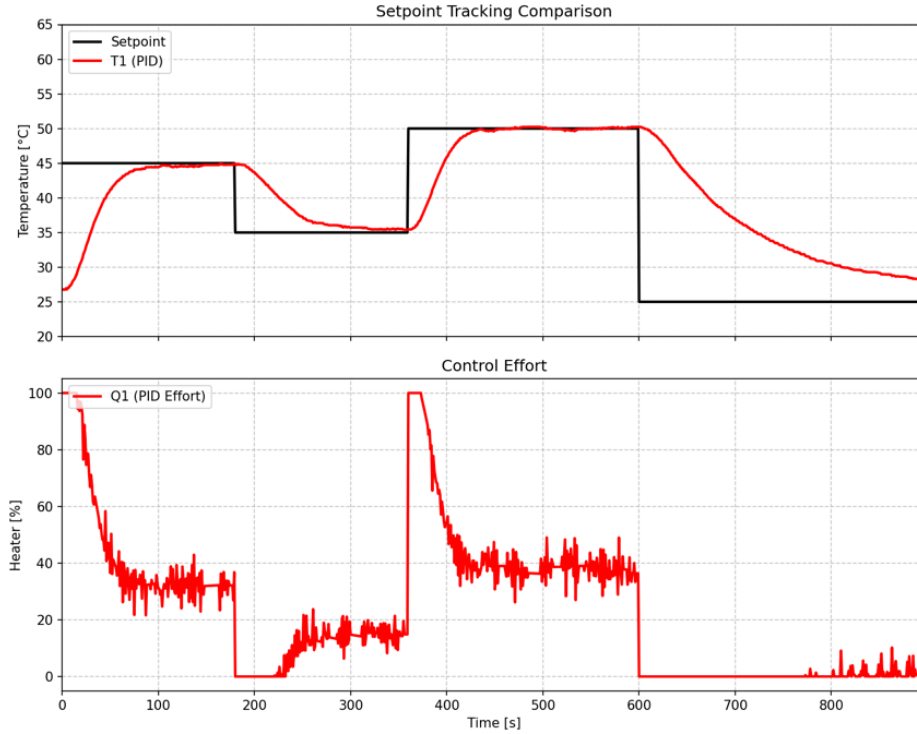


Figure 6: PID results.

Tuning procedure

The PID controller was tuned using the Skogestad Internal Model Control (SIMC) method for a First-Order Plus Dead-Time (FOPDT) model^[1], to obtain a compromise between speed, stability, and robustness. The identified plant parameters used in the tuning were mentioned above. The system has slow thermal dynamics and a measurable delay between heater actuation and sensor response, so the controller cannot be tuned too aggressively without risking overshoot or oscillatory behaviour.

The closed-loop time constant was selected as:

$$\tau_c = 1 * \theta$$

This tuning choice gives tight control, meaning the response is designed to be relatively fast while still maintaining robustness against the measured process dead time.

The controller parameters were calculated using the SIMC tuning equations:

$$K_c = \frac{1}{K} \frac{\tau}{\tau_c + \theta}; \tau_I = \min(\tau, 4(\tau_c + \theta)); \tau_D = \frac{\theta}{2}$$

This resulted in approximately $K_c = 7.466, \tau_I = 101.68\text{ s}, \tau_D = 6.355\text{ s}$

In the program, these values are converted into practical PID coefficients:

$$K_P = K_c; K_I = \frac{K_c}{\tau_I}; K_D = K_c \tau_D$$

A notable implementation detail is that the derivative contribution is computed from the measured temperature change rather than from the error derivative:

$$D = -K_D \frac{dPV}{dt} \neq K_D \frac{de}{dt}$$

This is especially relevant for this experiment because the setpoint profile contains sudden temperature jumps, and a derivative-on-error implementation would interpret those as extremely fast changes and would produce large artificial spikes in the controller output. However, the heater and sensor have not physically changed at the instant the setpoint changes, so derivative-on-measurement is more appropriate: it bases the damping action on the actual temperature trajectory of the system instead of on setpoint discontinuities. This improves actuator smoothness and avoids unnecessary heater chattering.

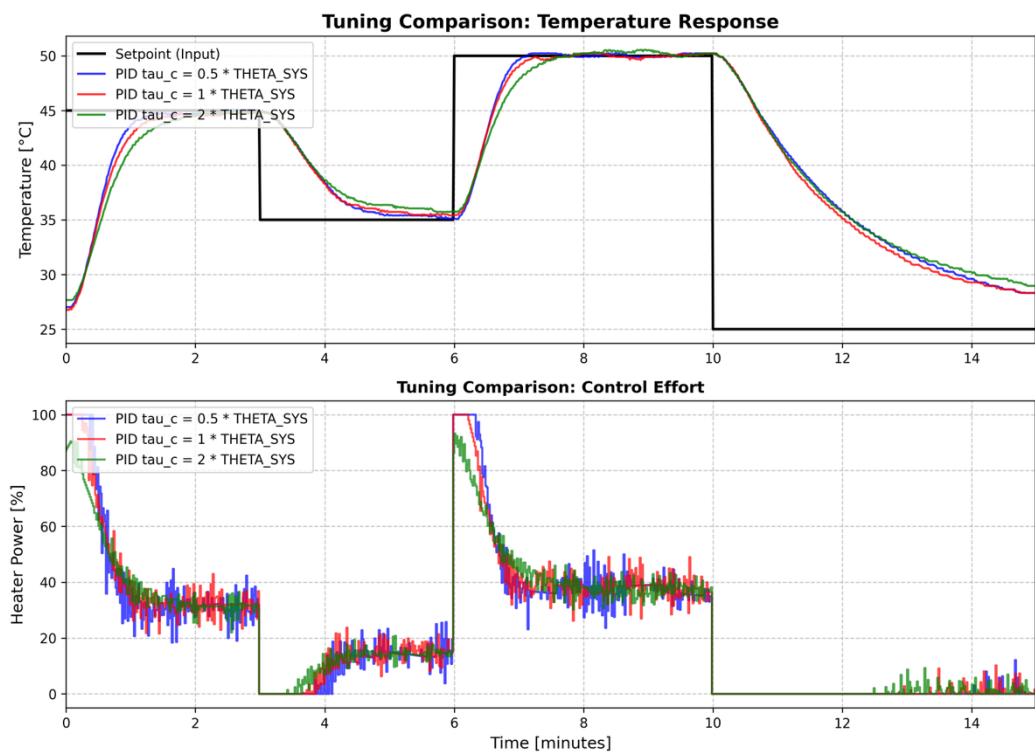


Figure 7: PID tuning.

TUNING PERFORMANCE METRICS COMPARISON			
Metric	PID tau_c = 0.5 * THETA_SYS	PID tau_c = 1 * THETA_SYS	PID tau_c = 2 * THETA_SYS
IAE (Tracking Error)	4877.56	4854.03	5237.76
ISE (Overshoot Pen.)	64687.15	62724.44	66035.80
ITAE (Settling Pen.)	2570319.26	2500157.40	2692613.30
Control TV (Wear)	3070.68	2770.41	2005.93

IAE: Integral of Absolute Error (Tracking Error) Measures how closely the controller follows the desired setpoint over time. It sums up the absolute deviation at every timestep. Lower values imply better tracking accuracy.

ISE: Integral of Squared Error (Overshoot Penalty) Similar to IAE but squares the errors, which heavily penalizes large deviations like overshoots. It's more sensitive to big spikes than small steady-state errors. Lower values mean less aggressive overshoot.

ITAE: Integral of Time-weighted Absolute Error (Settling Penalty) Weights errors by time, so errors that persist longer are penalized more severely. This specifically captures how quickly the system settles. Lower values are to be preferred for faster settling.

Control TV: Total Variation (Wear) Measures how much the control input moves around (sum of absolute changes in control signal). High TV means the actuator is working erratically, causing mechanical wear. Lower values lead to a smoother, less aggressive control.

Step 2 – System analysis and characterization

Main Dynamics and Delays

The system exhibits a dominant First-Order Plus Dead Time (FOPDT) dynamic. The slow thermal inertia of the physical components (the transistor acting as a heater and the surrounding air) dictates a large time constant ($\tau = 125.19\text{ s}$). Crucially, the spatial separation between the transistor and the temperature sensor introduces a pure transport delay ($\theta = 12.71\text{ s}$). This dead time is the most significant dynamic challenge of the system: it severely restricts the crossover frequency and fundamental closed-loop bandwidth. Any linear controller, such as the baseline PID, that attempts to force the system to react faster than this delay will inevitably cause severe overshoot or instability.

Nonlinearities and Asymmetric Dynamics

A major physical constraint of the plant is its asymmetrical dynamic response between heating and cooling phases. The actuator provides active input strictly in the positive direction, while the cooling phase is entirely passive, depending on natural convection and the ambient room temperature. This means the system can actively increase its temperature much faster than it can lower it.

Actuator Saturation and Windup

The system is subject to hard actuator saturation, with power physically constrained between 0% and 100%. When large setpoint changes occur, a purely linear controller calculates a theoretical control effort that far exceeds 100%. If not properly handled via anti-windup mechanisms, the integral term accumulates massive errors during saturation, leading to extreme overshoot and sluggish recovery.

External Disturbances

Finally, the plant is highly sensitive to unmeasured external disturbances, such as changes in ambient room temperature or localized airflow (drafts). Because the system relies on passive cooling, variations in convective heat transfer drastically alter the effective steady-state gain, requiring robust integral action to maintain zero steady-state error.

Step 3 – Controller design

Controller selection: MPC

For the advanced control strategy, a Custom Model Predictive Control (MPC) architecture was implemented from scratch utilizing the `scipy.optimize.minimize` algorithm (SLSQP method). The code is available as *MPC_heater.py*. MPC was selected over traditional PID variants because of its intrinsic ability to handle multi-

variable constraints and dead times. The TCLab heating process is characterized by a significant transport delay ($\theta = 12.71\text{ s}$); while the baseline PI controller reacts only after the error has materialized, the MPC utilizes the identified FOPDT internal model to predict future states over a receding horizon.

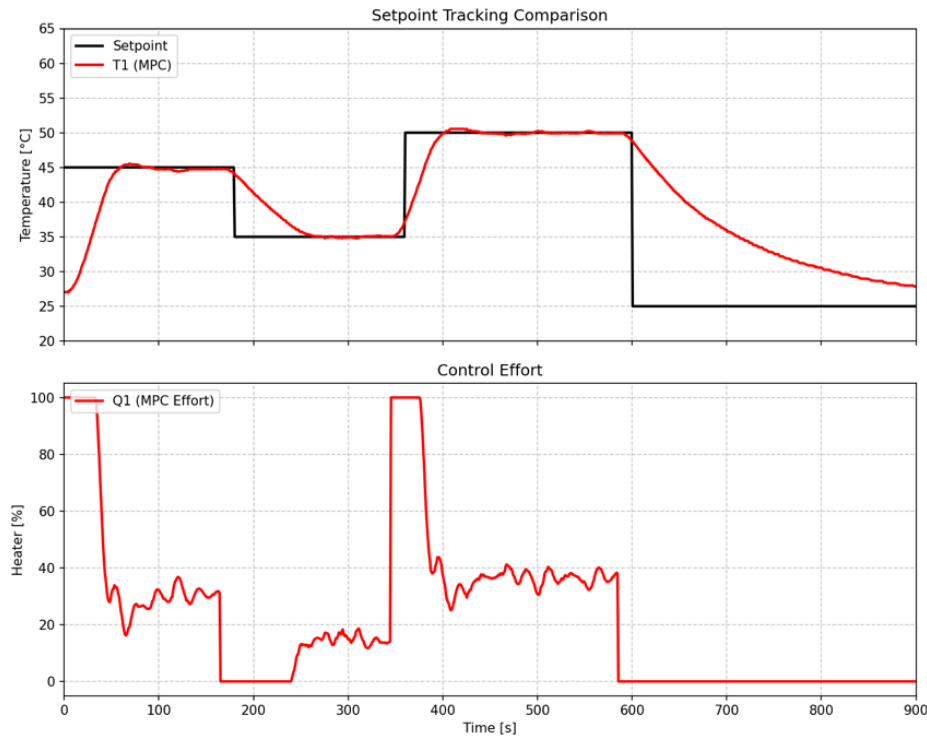


Figure 8: MPC results.

Nonlinearities and physical constraints

The physical heater subsystem exhibits two major operational constraints that challenge standard linear controllers: actuator saturation and asymmetric dynamics.

For the first problem, the MPC handles the boundaries directly in the optimization solver, preventing any mathematical windup.

As for the second, the MPC recognizes the physical limitation through the prediction horizon, aggressively cutting power to 0% in advance when downward setpoint steps are requested, allowing maximum time for passive thermal dissipation.

Controller tuning

The controller tuning was carefully balanced to prioritize smooth actuator tracking and computational stability:

- Prediction Horizon $N_p = 120$: set to evaluate the future trajectory over almost one time constant τ , ensuring the solver sees the full consequence of its immediate actions.
- Control Horizon $N_c = 25$: kept relatively short to enforce a conservative and computationally efficient control effort, preventing the solver from seeking overly aggressive, oscillatory solutions.
- Move Suppression $\lambda = 0.2$: penalty on the rate of change of the manipulated variable ΔQ was introduced to prevent actuator chattering, thus protecting the transistor hardware from thermal stress.
- Limited Foresight (15 seconds): slightly larger than the system's dead time, to prepare for the step change at the right moment, maximizing dynamic responsiveness while preserving predictive smoothness; a longer time would cause premature reactions.

To tune this controller, we worked on two levels: We first changed the full set of the three values $[N_p, N_c, \lambda]$, testing major differences. As a second step we tested the tuning for small scale changes, analysing the effect of the specific parameters. We then used the metrics to pick the preferred tuning set.

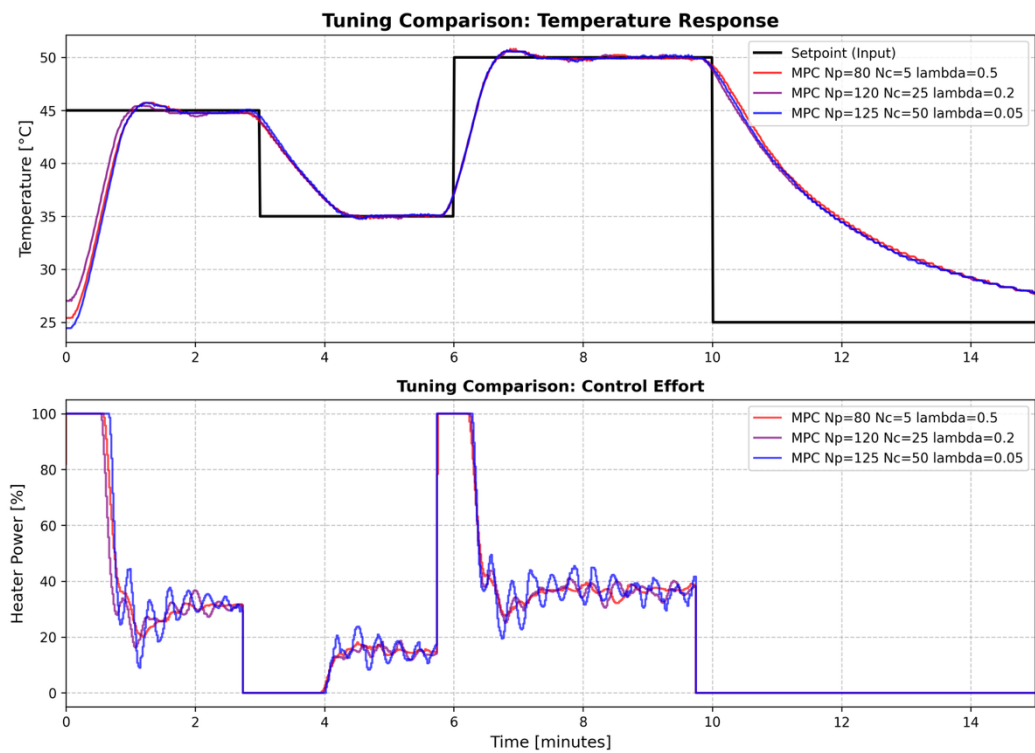


Figure 9: MPC large-scale tuning.

TUNING PERFORMANCE METRICS COMPARISON				
Metric	MPC $N_p=80$ $N_c=5$ $\lambda=0.5$	MPC $N_p=120$ $N_c=25$ $\lambda=0.2$	MPC $N_p=125$ $N_c=50$ $\lambda=0.05$	
IAE (Tracking Error)	4225.99	4031.24	4191.92	
ISE (Overshoot Pen.)	52864.94	48036.41	52208.67	
ITAE (Settling Pen.)	2230529.01	2161734.78	2177784.76	
Control TV (Wear)	441.76	551.44	866.32	

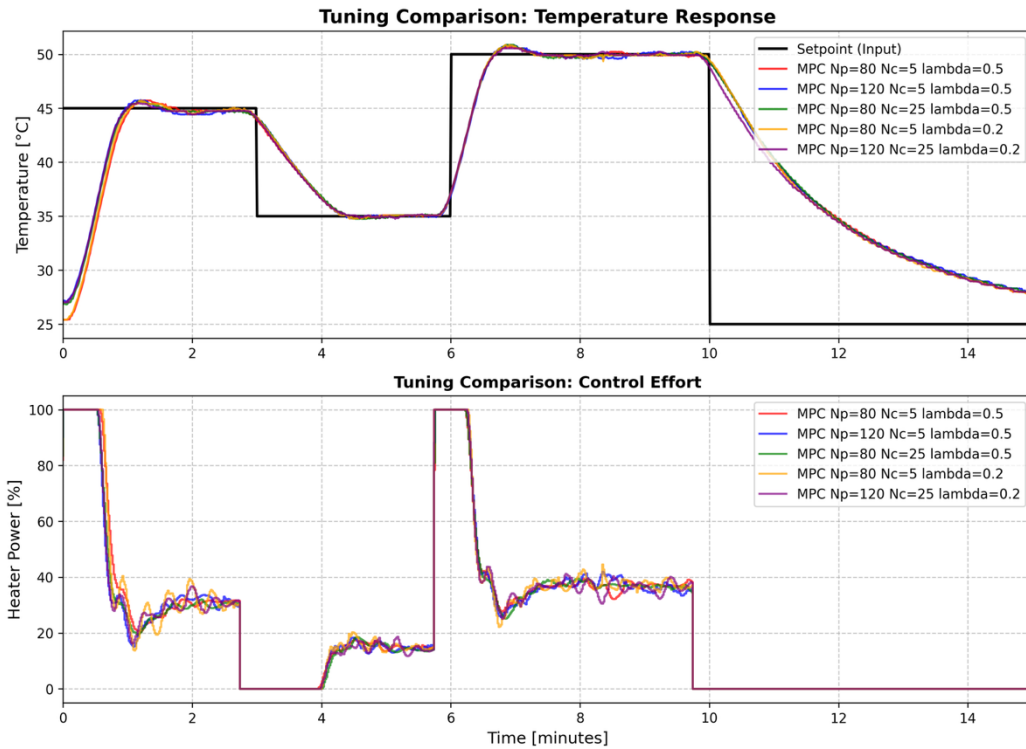


Figure 10: MPC small-scale tuning.

TUNING PERFORMANCE METRICS COMPARISON					
Metric Np=25 lambda=0.2	MPC Np=80 Nc=5 lambda=0.5	MPC Np=120 Nc=5 lambda=0.5	MPC Np=80 Nc=25 lambda=0.5	MPC Np=80 Nc=5 lambda=0.2	MPC Np=120 Nc=25 lambda=0.2
IAE (Tracking Error)	4225.99	4172.45	4165.03	4158.62	4031.24
ISE (Overshoot Pen.)	52864.94	50892.19	51257.39	51976.92	48036.41
ITAE (Settling Pen.)	2230529.01	2246287.03	2232877.78	2191989.51	2161734.78
Control TV (Wear)	441.76	508.79	434.19	634.92	551.44

Disturbance rejection

A known limitation of pure MPC algorithms is their susceptibility to steady-state offsets caused by unmeasured external disturbances (e.g., ambient airflow) or plant-model mismatch. To address this, an output disturbance estimator (bias updater) was integrated into the closed loop. At each sample interval, an internal shadow model simulates the theoretical open-loop temperature. The instantaneous difference between the measured sensor temperature and the theoretical model yields the bias term ($Bias = T_{measured} - T_{model}$). This bias is subsequently added to the entire future prediction horizon within the cost function. By dynamically shifting the prediction, the solver actively compensates for the missing heat, effectively providing the MPC with robust integral action that drives steady-state error precisely to zero.

To test that we operated the following experiment. The setpoint was fixed to 30°C for five minutes. From second 100 to second 130 the system was manually disturbed by blowing on it. From second 200 to second 230 it was disturbed by touching the heater's box with a hand. Here are the results:

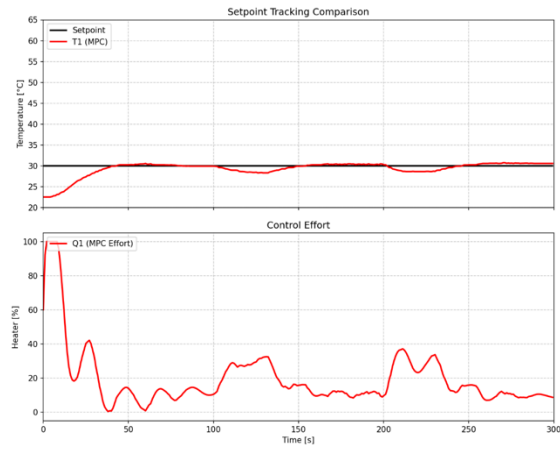


Figure 11: MPC disturbance test.

The disturbance was similar, even though in the second case the recovery is not as precise, which is probably since the surface that was touched changed temperature itself, and it took longer to go back to its previous temperature than for the heater alone. As we can see, the bias updater perfectly handles the disturbance issue.

Comparison

Both controllers successfully track the four setpoint steps throughout the experiment. However, clear differences emerge in the transient response and the control signal characteristics.

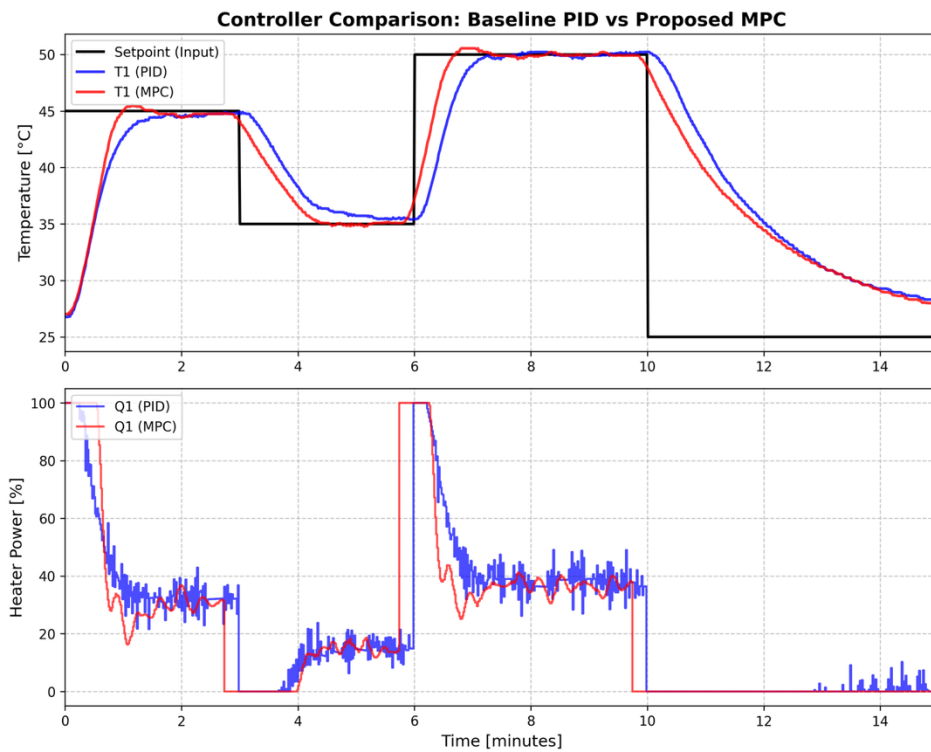


Figure 12: Controller comparison.

The **PID controller** exhibits a noisier control signal, with frequent and large oscillations in the heater power Q1, particularly during steady-state phases. This is a well-known consequence of the derivative action reacting to

measurement noise. Despite this, the temperature tracking remains acceptable, though with slightly slower settling and minor overshoots at certain steps.

The **MPC controller** produces a smoother control effort, since the optimizer explicitly penalises large changes in the control input within its cost function. The temperature response under MPC shows faster initial rise during upward steps (notably the step to 50 °C, where MPC reaches the setpoint ahead of PID) and a slightly cleaner approach to steady state.

During the cooling phase, both controllers behave similarly, since neither can actively cool the system: both simply cut heater power to zero and allow passive dissipation. It is also worth noting that during the cooling phase, the PID controller intermittently reactivates the heater with small power pulses in response to the diminishing error signal, partially counteracting the passive heat dissipation and causing the temperature to decrease more slowly than under MPC, which correctly holds the output at zero throughout.

Performance metrics

To provide a comprehensive and quantitative evaluation of the baseline PI and the proposed MPC controllers, we used the same performance metrics that were introduced previously, in the PID tuning section.

PERFORMANCE METRICS COMPARISON		
Metric	Baseline (PID)	Proposed (MPC)
IAE (Tracking Accuracy)	4854.03	4031.24
ISE (Overshoot Penalty)	62724.44	48036.41
ITAE (Settling Penalty)	2500157.40	2161734.78
Control TV (Actuator Wear)	2770.41	551.44

All four metrics favour the MPC controller. The most significant improvement is in Control Total Variation (TV), which quantifies the total variation in the control signal and directly relates to actuator wear. MPC reduces this by over 80%, reflecting its ability to plan smooth control actions over a prediction horizon rather than reacting instantaneously to the error signal. This is a fundamental advantage of model-predictive approaches over reactive controllers.

The IAE and ISE reductions confirm that MPC tracks the setpoint more accurately overall, largely due to faster rise times during upward steps. Interestingly, it is the MPC that exhibits slight overshoot in certain transitions, a consequence of its aggressive anticipatory action, while the PID approaches the setpoint more conservatively without overshooting. The ITAE improvement reflects that MPC, despite its occasional overshoot, settles closer to the setpoint for longer portions of the experiment, reducing the time-weighted accumulated error.

Limitations

Despite the clear performance advantages, MPC introduces additional complexity and computational overhead. The controller requires a process model (identified from step response experiments) and solves an optimisation problem at each sampling instant. If the model is inaccurate (due to nonlinearities, unmodelled dynamics, or environmental changes) MPC performance degrades. The PID, by contrast, is entirely model-free and robust to moderate model uncertainty by design.

Furthermore, the MPC parameters (prediction horizon, control horizon, and cost function weights) require careful tuning. An overly short horizon can cause myopic behaviour, while excessive penalisation of control effort may slow the response unacceptably.

Bibliography

[1] S. SKOGESTAD, "TUNING FOR SMOOTH PID CONTROL WITH ACCEPTABLE DISTURBANCE REJECTION, "INDUSTRIAL & ENGINEERING CHEMISTRY RESEARCH, VOL. 45, NO.23, PP. 7817–7822, 2006.